

CHEM 3A – Lab Practical - VERSION BSection #: Tues LabName: Ariel CeballosStudent ID #: 057637

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version B.

Station #1

mass of solute

$$\frac{50.0 \text{ mL}}{\text{mL solution}} \times 100 =$$

mL solution

Failed to write down all the information I needed to get the correct answer, didn't realise this until I had already left the station.

Station #2

empty vial = 17.2646g

Alum vial = 27.1943g

$$\begin{array}{r} 27.1943 \\ - 17.2646 \\ \hline \end{array}$$

9.9277g Alum

Actual yield

Station #3

44.1 mL $\text{H}_2(\text{g})$

Station #4

Temperature went up
reaction is exothermic

Station #5

$$3.7 \text{ cm} \times 10 = 37 \text{ mm}$$

Station #6

Observations: Fizzy at first then settled quickly

Reaction Type: Acid base neutralization

Balanced Equation: $\text{Na}_2\text{CO}_3 + \text{HCl} \rightarrow \text{Na}_2\text{CO}_3 + \text{HCl}$

Station #7

Volume withdrawn 10.00 mL

Initial Buret reading 1.27 mL

Final 17.12 mL

NaOH concentration 0.9573 M

$$M = \frac{\text{mol solute}}{\text{L solution}}$$

$$\frac{0.0004 \text{ M of citric acid}}{15.85 \text{ L}} = 0.9573 \text{ M}$$

Station #8

Test tube

Flask

Station #9

28 mL of water

25°C

Station #10

$$\frac{0.1003 \text{ g}}{0.2527 \text{ g}} \times 100 = 39.69\% \text{ of Oxygen}$$

Extra Space

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version C.

Station #1

$$5.0234 / 50.00 = \cancel{0.1004}$$

~~0.10~~ 0.10 M Solution A

Station #2

$$0.3298\text{g} / 474.46 \approx \textcircled{0.00069}$$

Station #3

$$22^\circ\text{C} + 756.9\text{mmHg} = \textcircled{778.9\text{mmHg}}$$

Station #4

~~35.0342g * 2.1°C~~ $q = m \cdot c \cdot \Delta T$
 $35.0342\text{g} \cdot 4.184\text{J/g}^\circ\text{C} \cdot 2.1^\circ\text{C} = 3043.344\text{J}$

Station #5

27mL displacement

3.4cm

Station #6

Observations: looks milky

Reaction Type: acid + base neutralization

Balanced Equation: $\text{BaPO}_4 \rightleftharpoons \text{Na}_2\text{I}_2$

Station #7

~~looks milky, ~~resistant~~ ; $\text{BaCl}_2 \rightarrow \text{Na}_2\text{PO}_4$~~
~~acid base~~ $\text{BaNa}_3 \rightarrow \text{Cl}_2\text{PO}_4$

19 mL

Station #8

Water glass, can't remember the name of the other piece of equipment but it's used in ~~exp~~ experiments to help measure pressure of gases

Station #9

3.9 cm

Station #10

~~$0.2479\text{g} / 0.1495\text{g} \cdot 100 =$~~

$0.1495\text{g} / 0.2479\text{g} \cdot 100 = 60.3065\%$

Extra Space

CHEM 3A – Lab Practical – VERSION A

Section #: _____

Name: Jasmine HernandezStudent ID #: 0871988

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version A.

Station #1

$$\frac{46.9067}{51.9344} = 0.9030g$$

density

Station #2

$$Alum = 8.879g$$

$$(24.98)(474.46) \times 100\% = 1280093.08\%$$

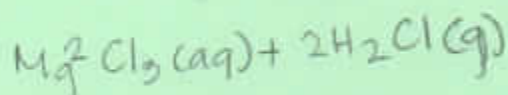
Percent yield of Reaction,
version A
7.4415g

there is not a reasonable value

Station #3

hydrogen gas

$$200.59 / 1.9 = 105.57g$$



Station #4

the cup was at 20°C

Station #5

$$V = l \times w \times h$$

$$3.6 \times 1.2 \times 1.2 = 5.184cm^3$$

Station #6

Observations: mixing the two caused a milky/cloud like

Reaction Type: ^{solution} precipitation

Balanced Equation: $\text{CaCl}_2 + \text{Na}_2\text{CO}_3$

Station #7

final Buret reading NaOH = 20 mL

$$\frac{6.9573 \text{ m}}{16.00 \text{ mL}} = 0.09573$$

Station #8

Flask	beaker	glass stand
burner	tube	pipet

Station #9

$$25.00 \text{ mL} / 2.5789 \times 100.00 \text{ mL} \\ = 0.644725$$

Station #10

$$\text{The mass of O is } 15.999 \text{ g} / 22.1722 \text{ g} \\ = 0.72159275 \text{ g} \\ \textcircled{0.72 \text{ g}}$$

Extra Space

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version A.

Station #1

$$T_{\text{Final}} - T_{\text{Initial}} = T_1 - T_2$$

17.0399 = volumetric flask + stopper 10.00 mL

$$\text{Density} = \frac{m}{V}$$

$$17.0399 = 17.04$$

$$\text{Density} = \frac{17.04 \text{ g}}{10.00} = 1.704 \text{ g/mL}$$

Station #2

$$\text{Molarity vs Molality} \rightarrow \frac{m}{L}, \frac{m}{kg}$$

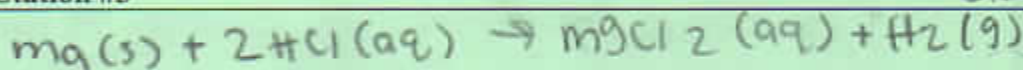
$$\text{Percent Yield} = \frac{\text{Actual}}{\text{theoretical}} = \frac{7.4415 \text{ g}}{8.879 \text{ g}} = .8381$$

.8381 $\times$ 100 = 83.81%, They were close to 100% meaning they done the experiment with less, there no excess or left over, otherwise the percentage couldve been over 100%.

Station #3

$$H = 1 \text{ mg} = 24.31 - 0.0475$$

$$Cl = 35.45$$



Station #4

The Temp of water (25 mL) was 23°C. and Stayed at 23°C.

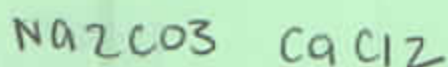
Station #5

Height = 3.5 cm

length = 1 cm

width = 1 cm

$$3.5 \text{ cm} \times 1 \text{ cm} \times 1 \text{ cm} = 3.5 \text{ cm}^3$$



Station #6

Observations: white, cloudy, chunky, separating

Reaction Type: Redox

Balanced Equation: $\text{Na}_2\text{CO}_3 + \text{CaCl}_2 \rightarrow 2\text{NaCl} + \text{CO}_3\text{Ca}$

Station #7

Final NaOH Buret = 20 mL

20 mL - 1.27 mL = 18.73 mL

0.9573

10 mL

$$\frac{0.9573}{18.73 \text{ mL}} = .0511$$

Station #8

- we have a bensen burner, used to connect tubes and heat stuff up.
- we have a 25 mL graduated cylinder.

Station #9

What is a molarity of a sol made by diluting 25.00 mL of 2.5789 M sodium chloride to a final buret reading of 100 mL?

$$\frac{2.5789 \text{ M}}{25.00 \text{ mL}} = .103156$$

$$.103156 / 100 \text{ mL} = .00103156$$

What mass of oxygen reacted?

Station #10

$$22.1715 - 21.8563 = .3152 \text{ g Mg} \times .04$$

$$22.1722 - 21.8563 = .3159 \text{ g Mg} \times .04$$

$$.3159 \text{ g} - .3152 \text{ g} = .0007 \text{ g}$$

Extra Space

CHEM 3A - Lab Practical - VERSION AName: Madi SantosSection #: version AStudent ID #: 0924739

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version A.

~~Station #1~~ Station #7

Final buret reading = 20 mL

~~V = 7.87~~~~V = 7.87~~V = 12.7

$$1.27 \times 10.00 = 12.7$$

$$V = \frac{10.00}{1.27} = 7.87$$

$$\frac{1.27}{10.00} = 0.127$$

~~Station #2~~ Station #8

A: graduated cylinder used to measure 20 mL of substance/liquid
 object attaches to how to transfer gas to start fire.
 oxygen

~~Station #3~~ Station #9

Molarity

2500 mL 2.57891 M NaCl

$$6022 - 2500? \times 100$$

24.088

Final volume of 100.00 mL

~~Station #4~~ Station #10

O = 15.999

empty crucible + lid = 21.8563 g

Mass = 0.1987g2.77g

crucible + lid + Mg = 21.9751g / 15.999 = 1.37

crucible + lid + Mg x O₂ = 22.1715, 114.3437 / 15.999 = 2.77?

0.0984 = 0.1987g
 0.1003
 22.1722 Mass of oxygen that reacted?

~~Station #5~~ Station #1Density = $\frac{\text{mass}}{\text{volume}}$

$$\frac{21.3712}{10} = 2.137$$

Density = 2.14

(M) Molarity = mol solute / L solution Mass volume percent $\frac{m}{V}\% = \frac{\text{mass solute g}}{\text{mL solution}} \times 100$

Station #6

Observations: Solution turned cloudy & white

Reaction Type: Redox

Balanced Equation: $\text{Na}_2\text{CO}_3 + \text{CaCl}_2 \rightarrow 2\text{NaCl} + \text{CaCO}_3$

~~Station #7~~ station #2

Percent yield (%) = (actual yield) / (theoretical yield) $\times 100\%$

8.879g ← theoretical

7.4415g ← actual?

$$7.4415 / 8.879 \times 100 = 83.81\%$$

~~Station #8~~ station #3

Mass of hydrogen gas using 0.0475g piece of Mg

1.008 = H

24.31 = Mg

$$24.31 \times 0.0475 = 1.155\text{g}$$

~~Station #9~~ station #4

temp of water in styrofoam cup = 22.5°C

~~Station #10~~ station #5

$$V = L \times W \times h$$

cm^3

L = 3.6cm

W = 1.2cm

h = 1.2cm

$$3.6 \times 1.2 \times 1.2 = 5.184\text{cm}^3$$

Extra Space

Station #6

Observations: cloudy, white, no saturation, combination

Reaction Type: ~~no reaction~~ A/B neutralization

Balanced Equation: $\frac{1}{1} \text{Na}_3\text{PO}_4 + \frac{1}{1} \text{BaCl}_2 \Rightarrow \frac{1}{1} \text{Na}_3\text{Ba} + \frac{1}{1} \text{PO}_4\text{Cl}_2$

Station #7

Final Reading 20.0 mL

0.9573 M

Station #8

Watch glass & Volumetric Pipette

1
I/O/C how to
SPILL IT

Station #9

31.82 cm

Station #10 Start Point

MASS PERCENT OR MG OR MG OX

~~9.1810%~~

65.182%

MASS USED
1.1445 g
MG OX 1.2479 g
ORIGINAL MASS 0.10424 g

Extra Space

CHEM 3A - Lab Practical - VERSION C

Section #: _____

Name: Sergio GomezStudent ID #: 1039936

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version C.

Station #1

$$5.0234 \text{ g NaCl} \times \frac{1 \text{ mol}}{58.44277 \text{ g NaCl}} \times \frac{1000 \text{ mL}}{50.00 \text{ mL}} \times \frac{50.00 \text{ mL solution}}{5.0234 \text{ g NaCl}}$$

calculate molarity
molar mass

$$117191 \text{ mol/L}$$

Station #2

$$13302.9 \text{ g Al} \times \frac{1 \text{ mol Al}}{26.98 \text{ g Al}} \times \frac{2 \text{ mol Al}}{2 \text{ mol Al}} \times \frac{474.146 \text{ g Al}}{1 \text{ mol Al}} = 580.5 \text{ g Al}$$

Alum = 474.146 g/mol
mass of foil = 13302.9 g
which amount of alum can be made

Station #3

$$22.0^\circ\text{C} - \text{Temp}$$

$$P_{\text{total in glass}} = 756.6$$

pressure of H_2 in mmHg
can't do w/o Boyle's Law

$$P_{\text{total}} = 756.6 +$$

Station #4

$$3093.3 \text{ J} \leftarrow \text{heat absorbed positive \#}$$

$$q = m \times C \times \Delta T$$

mass of water 35.0392 g
 $C = 4.184 \text{ J/g}^\circ\text{C}$
 $\Delta T = 21.1^\circ\text{C}$

Station #5

$$22.8 \text{ mL}$$

volume of water

CHEM 3A – Lab Practical - VERSION B

Section #: _____

Name: Samaria NamvilaythongStudent ID #: 1012960

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version B.

Station #1

$\frac{100.1147g}{50.0mL} = 2.002294 g/mL \times 100$ $\frac{m}{V} \% = 200.2294 g/mL$	mass vol $\frac{m}{V} \% =$ 50 mL flask + stopper = 50 flask + stopper + NaCl flask + stopper + solution	mass solution ML solution: $\times 100$ 46.9067g 51.9344g 100.1147g
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Station #2

mass of alum 27.1436g 17.2659g $\leftarrow$ empty mass of alum	Actual yield
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Station #3

Volume of gas = 42.8 mL

Station #4

20°C $\rightarrow$ 19°C

endo or exo exothermic

Station #5

2.8cm = 12in millimeters

Object B = 3.9cm mm?

$$\frac{3.9cm}{1cm} \times \frac{1000mm}{1cm} = 3900mm$$

Station #6

Observations: Clear, bubbled when $(\text{Na}_2\text{CO}_3 + \text{HCl})$ interacted, cold

Reaction Type: Acid/Base Neutralization

Balanced Equation: $2\text{Na}_2\text{CO}_3 + \text{HCl} \rightarrow 4\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$

Station #7

$$M = \frac{\text{moles solute}}{\text{L solution}}$$

$$M = 0.01083 \text{ M}$$

Vol 6 Down Soda = 10.00 mL
 In Buret = 1.27 mL
 F Buret = 17.12 mL
 15.85 mL

$$\frac{15.85 \text{ mL}}{1000 \times 10} = 16$$

Station #8

- test tube
- flask volumetric = measures volume of substances

Station #9

Volume of water in graduated cylinder = 27.8 mL
 temp. of water in flask = 24.9 °C

Station #10 0.1524 g Mg = 24.305 O = 15.999

1. mass of Mg used:
 2. Mass of Mg_xO_y compound: 0.2527 g
 3. Oxygen Mg_xO_y compound: 0.1003 g
- Mass percent of oxygen in Mg_xO_y sample

$$\frac{0.1003 \text{ g}}{0.2527 \text{ g}} \times 100 = 39.71\%$$
- 16% O in Mg_xO_y sample

Extra Space

CHEM 3A - Lab Practical - VERSION B

Name:

Juana Lopez

Section #: 2, Tuesday

Student ID #:

1028207

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version B.

Station #1

mass - volume % of solution A

$$\frac{51.9344 \text{ g}}{100.1197 \text{ g}} = 51.87489949$$

$$51.8748 \text{ g}$$

Station #2

- mass Alum in vial B: 27.1939g

- Empty vial: 17.2666g

This measurement is the actual yield.

Station #3

- 44.03 mL 756.5

- 2P - 19.650

gases collected ~

Station #4

initial temp. 23

final temp. 20

exothermic.

Station #5

3.6 cm → 5.76 mm

Station #6

Observations: I used a test tube full up 1 mL of Na_2CO_3 & HCl nothing happened NO reaction

Reaction Type: $\text{Na}_2\text{CO}_3 / \text{HCl} \rightarrow \text{Na}_2\text{CO}_3 \rightarrow \text{HCl}$

Balanced Equation: NO reaction

Station #7

Volume of "G Down" soda - 10.00 mL

Initial Buret reading (NaOH) - 1.27 mL

Final Buret reading

If it was 17.12 mL molarity (M) =

Station #8

- 125 mL flask measurement

- testing tube

Station #9

- measure of cylinder is: 28.0

- 25°C

Station #10

1. mass of Mg used: 0.1524 g & mass % used of oxygen in MgO is = 65%
2. mass of the Mg: 0.2527 g
3. mass of oxygen in MgO compound: 0.1003 g

Extra Space

CHEM 3A - Lab Practical - VERSION CSection #: Chem 3AName: Rebecca WrightStudent ID #: 0934216

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version C.

Station #1

50.00 mL sol'n A
 5.0234 g NaCl
 calculate molarity of solution A
 MM = 58.44277 g/mol

$$50.00 \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}} = 0.05 \text{ L}$$

$$\frac{5.0234 \text{ g}}{58.44 \text{ g/mol}} = 0.0860 \text{ mol}$$

$$\frac{0.0860 \text{ mol}}{0.05 \text{ L}} = \boxed{1.72 \text{ M}}$$

molarity(M) = $\frac{\text{mol solute}}{\text{L solution}}$

Station #2

$$\text{mass} = 0.3301 \text{ g Al}$$

$$\boxed{0.7692 \text{ g}}$$

Station #3

Water temp = $21.8^\circ\text{C} + 273 = 294.8 \text{ K}$
 Pressure = 19.587 mmHg
 $\text{H}_2\text{O} = 44 \text{ mL}$
 $P_{\text{total}} = P_{\text{H}_2} + P_{\text{H}_2\text{O}}$

$$P_{\text{H}_2} = \frac{19.587}{294.8} = 0.06644$$

$$\boxed{P_{\text{H}_2} = 0.06644}$$

Station #4

$$35.0392 \text{ g} \times 4.184 \text{ J/g}^\circ\text{C} \times 21.1^\circ\text{C}$$

$$= \boxed{3093}$$

Station #5

$V = 21.1 \text{ mL}$
 $m = 3.9 \text{ cm}^3$

$$\frac{21.1 \text{ mL}}{3.9 \text{ cm}^3} \times \frac{1 \text{ cm}^3}{1} = \boxed{5.4 \text{ mL}}$$

Station #6

Observations: became cloudy/milky when combined

Reaction Type: Acid Base Neutralization

Balanced Equation: $\text{BaCl}_2(s) + \text{Na}_3\text{PO}_4(aq) \rightarrow \text{BaCl}_3\text{Na}_3\text{PO}_3$

Station #7

$$10.00\text{mL} + 1.27\text{mL} = \boxed{11.27\text{mL}}$$

Station #8

watch glass
buret

Station #9

3.9 cm

Station #10

mass of Mg used: 0.1495g

mass $\text{Mg} \times \text{O}_y$: 0.2479g

mass O_2 in $\text{Mg} \times \text{O}_y$: 0.0984g

$$\frac{0.2479\text{g Mg} \times \text{O}_y}{0.1495\text{g Mg}} \times 100 = \boxed{60.31\%}$$

Extra Space

CHEM 3A – Lab Practical - VERSION BName: Danich Rors

Section #: _____

Student ID #: 096444

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version B.

Station #1

46.9067g
51.9344g
100.1147g

51.9344g
- 49.9067g

2.0277g

$\frac{2.0277g}{20.0\text{ mL}} \times 100 = 10.0554\%$

Station #2

Empty - 17.2662g
Full - 27.1931g

$27.1931g - 17.2662g = 9.9269g$

Station #3

21° 44

Station #4

21° → 22°

Exothermic

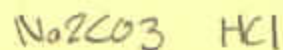
Station #5

3.9 cm → 39 mm

$3.9 \times 10 = 39\text{ mm}$

Station #6

Observations: fizzled then no reaction



Reaction Type: No reaction

Balanced Equation: $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$

Station #7

20 mL

Station #8

Erlenmeyer flask
test tube

Station #9

28.5 mL

25°C

Station #10

$$\frac{0.1003\text{g}}{0.2527\text{g}} \times 100 = 39.7\%$$

Extra Space

CHEM 3A – Lab Practical - VERSION BSection #: 43958Name: Joscelyne OleaStudent ID #: 619422

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version B.

Station #1

$$100.1147g - 46.9067g = \frac{53.208g}{50.00}$$

46.9067

$$51.9344 - 46.9067g = \frac{5.0277g}{50.00mL} \times 100 = 10.06\%$$

Station #2

empty: 17.2664g

$$27.1937g - 17.2664g = 9.9273g \text{ Alum}$$

Filled: 27.1937g

Theoretical yield

Station #3

43.73
~21.10°

19.113 mmHg

Station #4

- Temp went down
- Endothermic

Station #5

$$3.8cm \times 10^{-3} = 0.0038 mm$$

Station #6

Observations: It bubbled, then stopped



Reaction Type: Acid/Base neutralization



Balanced Equation:

Station #7

$$\frac{39.957}{15.85 \text{ mL}} = 0.0025 \text{ M ?}$$

start
▶

Station #8

- Erlenmeyer flask (125): can be used to hold or measure solutions
- Test tube: To hold / mix solutions

**Station #9**

Volume of water $\approx 27.8 \text{ mL}$

Temp $\approx 24.5^\circ\text{C}$

Station #10

$$\frac{0.1003}{0.2527} \times 100 = \underline{39.69\%}$$

Extra Space

CHEM 3A - Lab Practical - VERSION CName: Evelyn Muñoz

Section #: _____

Student ID #: 1047765

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version C.

Station #1

$$= 5.0234 / 50.00$$

$$0.100468 \text{ g/mol}$$

$$A = 50.00 \text{ mL}$$

$$5.0234 \text{ g NaCl}$$

molar mass:

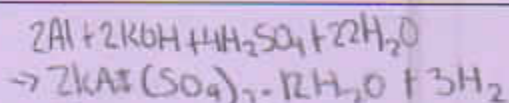
$$58.4427 \text{ g/mol}$$

$$m = \text{mol solute} / \text{L solution}$$

Station #2

weight of alum: 0.3302 g

$$0.3302 / 26.98$$

$$0.01224 \text{ grams can be made}$$


Substance	molar mass
alum foil (Al)	26.98 g/mol
Alum ($\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$)	474.46 g/mol

Station #3

$$22^\circ\text{C}$$

$$P_{\text{total}} = P_{\text{H}_2} + P_{\text{H}_2\text{O}}$$

$$756.8 + 22 = 778.8 \text{ mm Hg}$$

Station #4

mass of NaOH	30.25 g
mass total solution (m)	35.6392 g
Heat of Solution (C)	4.184 J/g°C
change in temp. (ΔT)	21.1°C

$$q = m \times C \times \Delta T$$

$$35.6392 \times 4.184 \times 21.1$$

$$q = 3093.345$$

Station #5

$$\text{Initial DI: } 14 \text{ mL}$$

When the atom was placed in the distilled water it moved up to 21 mL meaning it went up

$$21 \text{ mL}$$

$$7 \text{ mL}$$

Station #6

Observations: When BaCl_2 and Na_3PO_4 were mixed it turned cloudy white

Reaction Type: Acid / Base Neutralization

Balanced Equation:

Station #7

NaOH concentration 0.9573 M

Volume 10.00 mL

Initial 1.27 mL

$$0.9573 \times 1.27 / 10.00$$

Final
reading

$$= 0.1216 \text{ mL}$$

Station #8

glass watch

volumetric buret

Station #9

3.9 cm

Station #10

Mg used: 0.1495 g

Mg₂O₃ compound: 0.2479 g

Oxygen in Mg₂O₃ compound: 0.0984 g

$$0.1495 / 0.2479 = 0.6031 = 60.31\%$$

$$0.006031\%$$

Extra Space

CHEM 3A - Lab Practical - VERSION B

Section #: _____

Name: Teah PrescaleStudent ID #: 1035199

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version B.

Station #1

$$\text{mass vol\%} = \frac{\text{mass solute}}{\text{vol solution}} \times 100$$

$\text{flask} = 46.2667\text{g}$
 $\text{flask + steel} = 51.0344\text{g}$
 $\text{flask + solution} = 100.1147\text{g}$
 $\frac{4.9677\text{g}}{52.148\text{g}} \times 100 = 9.34601304\%$
9.346% or 9.35%

Station #2

$$\text{Vial + alum} = 27.1638\text{g}$$

$$\text{vial} = 17.2665\text{g}$$

$$27.1638\text{g} - 17.2665\text{g} = 9.8973\text{g}$$

this is the actual yield.

Station #3

record vol of sol in graduated cylinder

mg (s)	12401 (aq)	44913 (aq)	162 (g)
400 mL	43 mL	22°C	19.688g
19.327	20.070	20.31	20.585

Station #4

$22^\circ \rightarrow 21^\circ$
 went down
 endo thermal
 created its own temp
 once soln added

Station #5

3.8 cm
 $3.8\text{ cm} = 38\text{ mm}$
 $\text{cm} = 10^{-2}$
 $\text{mm} = 10^{-3}$

Station #6

Observations: Both inside defect but no other reactions. Product looks like oil & water can see flask swirling

Reaction Type: ~~1st reaction~~ ^{acid base} $A + B = AB$

Balanced Equation: $\text{Na}_2\text{CO}_3 + \text{HCl} \rightarrow \text{Na}_2\text{CO}_3 + \text{HCl}$

Station #7

17.12 mL 10.00 mL

$M_a = 22.990$ $a = 15.999$ $M = 1.6579$ $a = 39.9909 = \text{NaOH}$

$17.12 \text{ mL} - 1.27 \text{ mL} = 15.85 \text{ mL}$

0.2573 M

0.1655698318

$M = \text{mol solute} / \text{L solution}$

0.16557 M

Station #8

Erlenmeyer flask - used to mix solutions when doing reactions

Test tube - used to mix substances together to see their results.

Station #9

20.8 mL in graduated cylinder

25°C

Station #10

$m_1 = 0.1524 \text{ g}$

$m_{\text{vol}} = 0.2527 \text{ g}$

65.81%

$0.1524 \text{ g} / 0.2527 \text{ g} = 0.6034$

$\frac{0.1524 \text{ g}}{0.2527 \text{ g}} \times 100 =$

Extra Space

CHEM 3A – Lab Practical - VERSION C

Section #: _____

Name: Kristie CorcaStudent ID #: 0994114

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version C.

Station #1

$$\text{Molarity} = 5.0235 / 50 \text{ mL} = 0.10047 \text{ M}$$

Station #2

$$\text{mass} = 0.3309 \text{ g}$$

$$= 1.50$$

$$= 0.00220416$$

Station #3

$$\text{Temp of water } 21.5^\circ\text{C}$$

$$P_{\text{Total}} = 756.7$$

$$756.7 = 400 \text{ mL} + 713.9$$

$$42.8$$

Station #4

$$30.93$$

Heat q

$$30.25$$

$$m = 35.0392 \text{ g}$$

$$(c) = 4.184 \text{ J/g}^\circ\text{C}$$

$$\Delta T = 21.1^\circ\text{C}$$

$$q = m \times c \times \Delta T$$

$$35.0392 \times 4.184 \times 21.1$$

$$35.0392 \times 21.1$$

Station #5

400 mL beaker

filled w/ water 275 mL

125 mL

285 mL w/ object

10 mL displaced

Station #6

Observations: White, foggy.

Reaction Type: Acid/Base neutralization

Balanced Equation: $\text{BaCl}_2 + \text{Na}_3\text{PO}_4 \rightarrow \text{BaCl}_3\text{Na}_3\text{PO}_3$

Station #7

Final Buret reading: 20.1 ml

Station #8

Watch glass 3rd item is a 1cm dropper used to transfer, add in liquids during a lab.

Station #9

3.9 cm

Station #10

1. $0.1495 / 24.305 = 0.00615$

2. $0.2479 / 15.999 = 0.01549$

3. $0.0984 / 0.01549 = 6.3525 \times 100$

Mass % = 64%

Extra Space

CHEM 3A – Lab Practical – VERSION ASection #: 43957Name: Tricia SnellStudent ID #: 1068669

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version A.

Station #1

mass volumetric flask = 13.0872 g
 mass volumetric flask + soln = 23.5887 g
 mass of soln = 10.5015 g
 volume of soln = 10 mL

$$d = \frac{10.5015 \text{ g}}{10 \text{ mL}} \times \frac{1 \text{ mL}}{1 \text{ cm}^3} = \frac{10.5015 \text{ g}}{10 \text{ cm}^3} = \boxed{1.05015 \frac{\text{g}}{\text{cm}^3}}$$

Station #2

Th = 8.879 g
 Ac = 7.4415 g

$$\frac{7.4415 \text{ g}}{8.879 \text{ g}} \times 100 = 83.81\% \text{ percent yield}$$

= reasonable value

Station #3

$\text{Mg}(s) + 2\text{HCl}(aq) \rightarrow \text{MgCl}_2(aq) + \text{H}_2(g)$
 H in g = ?
 0.0475 g Mg
 Mg = 24.31 g/mol H = 1.008 g/mol

$$0.0475 \text{ g Mg} \times \frac{1 \text{ mol}}{24.31 \text{ g/mol}} \times \frac{1 \text{ mol Mg}}{2 \text{ mol H}} \times \frac{1 \text{ mol H}}{1.008 \text{ g}} = 9.69 \times 10^{-4} \text{ g H}$$

$$\frac{0.0475}{49.00896} = 0.000969211 = \boxed{9.69 \times 10^{-4} \text{ g H}}$$

Station #4

Temp of 25 mL water in plastic cup (in styrofoam cup) = $\boxed{22.90^\circ \text{C}}$

Station #5

L = 3.70 cm
 W = 1.30 cm
 H = 1.20 cm

$$V = l \times w \times h$$

$$V = 3.70 \text{ cm} \times 1.30 \text{ cm} \times 1.20 \text{ cm}$$

$$V = 5.772 \text{ cm}^3 = \boxed{5.77 \text{ cm}^3}$$

Station #6

Observations: solution turned white, some precipitate formed

Reaction Type: Precipitation

Balanced Equation: $\text{Na}_2\text{CO}_3 + \text{CaCl}_2 \rightarrow 2\text{NaCl} + \text{CaCO}_3$

Station #7

Final Reading: 20.05 mL NaOH

Initial reading: 1.27 mL

Volume used: $20.05 \text{ mL} - 1.27 \text{ mL} = 18.78 \text{ mL of NaOH used}$

Station #8

1) graduated cylinder

2) bunsen burner

Station #9

$V_1 = 25.00 \text{ mL}$

$M_1 = 2.5789 \text{ M NaCl}$

$V_2 = 100.00 \text{ mL}$

$M_2 = ?$

$$25.00 \text{ mL} \times 2.5789 \text{ M} = 100.00 \text{ mL} \times M_2$$

$$\frac{64.4725 \text{ mL M}}{100.00 \text{ mL}} = 0.644725 \text{ M} = \boxed{0.6447 \text{ M}}$$

Station #10

crucible + lid = 21.8563 g

crucible + lid + Mg = 21.9751 g

crucible + lid + Mg, O₂ = 22.1715 g (1)

22.1722 g (2)

mass of O reacted = ?

$$22.1722 \text{ g} - 21.9751 \text{ g} = \boxed{0.1971 \text{ g of O reacted}}$$

Extra Space

CHEM 3A – Lab Practical – VERSION A

Section #: _____

Name: Emily S.Student ID #: 1022282

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version A.

Station #1

$$D = \underline{20.0503g}$$

Station #2

$$838\% = \frac{7.4415g}{8.879g} \times 100\%$$

Station #3

Station #4

$$21^{\circ}C$$

Station #5

$$V = l \times w \times h$$
$$3.5cm^3 = 3.5cm \times 1cm \times 1cm$$

Station #6

Observations: It was milky looking, looked as if there was separation.

Reaction Type: Acid base neutralization

Balanced Equation:

Station #7

19.90 mL = NaOH final reading

8.73 mL of NaOH was used

Station #8

1 = graduated cylinder (25 mL)

2 = Bunsen burner

Station #9**Station #10**

0.0971 was the mass
of oxygen that reacted.

$$\begin{array}{r} 28.7722 \\ - 21.9751 \\ \hline 0.0971 \end{array}$$

Extra Space

CHEM 3A – Lab Practical – VERSION A

Section #: _____

Name: Beatrice EsparzaStudent ID #: 0888543

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version A.

Station #1

10.00 mL volumetric flask
calculate density of solution 20.0932 g
mass of flask 9.6870 g
Density = $\frac{\text{mass}}{\text{volume}}$ 0.482103398

Station #2

most Alum calculated made 8.879 g.
Alum made was 7.4415 g.

Percent yield(%) = (Actual Yield)/(Theoretical Yield) $\times$ 100%
is this a reasonable value?

Station #3

$\text{Mg(s)} + 2\text{HCl(aq)} \rightarrow \text{MgCl}_2\text{(aq)} + \text{H}_2\text{(g)}$
calculated mass of hydrogen gas (in grams) if used a
0.0475 g piece of magnesium

Station #4 of water

25 mL added to plastic cup then placed in styrofoam cup
23°C in temperature

Station #5

Determine the volume of object by measuring dimensions
Answer in cm^3

Length = 3.7 cm

width = 1.2 cm

volume = 4.44 g.

$V = l \times w \times h$

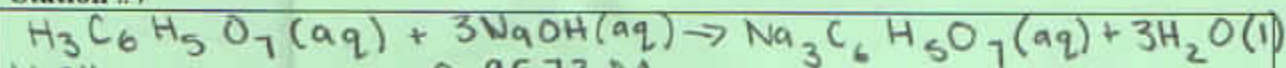
Station #6

Observations: turned milky color

1 mL of 2 Solutions
Added

Reaction Type: Acid/Base neutralization

Balanced Equation:

Station #7

NaOH concentration 0.9573 M

volume of Soda = 10.00 mL

initial Buret reading NaOH = 1.27 mL

Final buret Reading =

Station #8

graduated Cylinder

Station #9

final molarity 25.00 mL of a 2.5789 M Sodium chloride solution
to a final volume of 100.00 mL

Station #10

0.1524

0.2527

0.1003

g 0.603 mass of Mg used

g 0.8587 mass of Mg_xO_y compoundg 0.956 mass of Oxygen in Mg_xO_y compound**Extra Space**

CHEM 3A – Lab Practical - VERSION B

Section #: _____

Name: Serena PedrozaStudent ID #: 0995473

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version B.

Station #1

44.9067

$$\frac{51.9344\text{g}}{50.00\text{mL}} \times 100$$

Station #2

Actual Yield

17.2671g
27.1940g

Station #3

Record V of gas

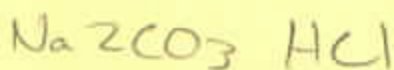
H₂(g) 44.9 mL

Station #4

- Temp went down
- Endothermic

Station #5

372 cm = Convert to mm



Station #6

Observations:

Small bubbles when first mixed

Reaction Type:

No reaction

Balanced Equation:

?

Station #7

0.9573 M

1.27 mL

10.00 mL

Molar mass of NaOH, initial V of NaOH, & Vol of Soda

Final = 17.12 mL what is molarity of $\text{CH}_3(\text{C}_6\text{H}_5\text{O}_2)$ in the "6 down" soda?

$$10.00 \text{ mL} - 17.12 \text{ mL} = 7.12 / 29.0179$$

$$0.24536 = 0.2454 \text{ M}$$

29.0179

Station #8

- Test tube

- Erlenmeyer Flask

Station #9

Vol. of water = 28.0 mL

Temp = 24.8°C

Station #10

Mass Mg = 0.1524 g

MgO = 0.2527 g

O = 0.1003 g

Mass Percent

$$= \frac{0.1003}{0.2527} \times 100$$

39.69% 39.69133

Extra Space

CHEM 3A – Lab Practical - VERSION C

Section #: _____

Name: Andres GarciaStudent ID #: 0790131

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version C.

Station #1

$$\frac{5.0234}{50.00 \text{ mL}} = 0.100$$

to make 50.00 mL of Sol A

$$5.0234 \text{ g of NaCl}$$

$$\text{molar mass of NaCl} = 58.44277 \text{ g/mol}$$

50.00

$$\text{Molarity} = (\text{mol solute}) / (\text{L solution}) \quad 5.871 = M$$

Station #2

$$\text{mass of C} = 0.3306 \text{ g}$$

0.0122

$$\text{Al} = 26.98 \text{ g/mol}$$

$$\text{Alum} = 474.46 \text{ g/mol}$$

$$0.000025826 \text{ g of Alum}$$

Station #3

$$21^\circ\text{C} = 1$$

$$756.9 \text{ total} = 2$$

$$p_{\text{total}} = p_{\text{H}_2} + p_{\text{H}_2\text{O}}$$

what is
pressure of
H₂ gas in collection
tube

$$43.5 \text{ mL}$$

$$18.650 \text{ mm Hg}$$

$$3 = 738.25 \text{ mm Hg}$$

$$756.9 = p_{\text{H}_2} + 18.650$$

$$- 18.650 \quad - 18.650$$

Station #4

$$q = m \times C \times \Delta T$$

heat released

$$35.0392 \text{ g} \times 4.184 \text{ J/g} \times 21.1^\circ\text{C}$$

$$21.1 + 273.15$$

$$294.25 \text{ K}$$

$$q = 43138.23 \text{ J}$$

$$43.138 \text{ kJ}$$

Station #5

$$3.8 \text{ mL}$$

$$\text{Starting} = 20 \text{ mL}$$

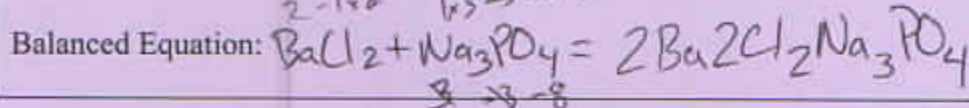
$$\text{ending} = 27 \text{ mL}$$

$$\text{Volume of C} = 7 \text{ mL}$$

Station #6

Observations: Very cloudy, milky white

Reaction Type: Redox

**Station #7**

20.1 mL

Station #8

- Volumetric pipet - used for very accurate ~~measurements~~ measurements to the .01 mL
- Weighing glass - used to weigh product on the equipment

Station #9

3.9 cm

Station #10

$$\frac{0.1495}{0.2471} = 60.3\% \text{ of magnesium in the } \text{Mg}_x\text{O}_y \text{ sample}$$

Extra Space

CHEM 3A – Lab Practical – VERSION ASection #: TUE 120Name: Lisa VargasStudent ID #: 0775232

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version A.

Station #1

$$\frac{9.5863}{13.95767} \times 100 = 68.68123405\%$$

%68

Station #2

$$\frac{8.879\text{g}}{25.3797\text{g}} \times 100 = 34.98465309\%$$

Station #3

$$24.31 + 4003 + 3545 = 68.763\text{g}$$

Station #4

$$23^{\circ}\text{C}$$

Station #5

$$6.084\text{ cm}^3$$

Station #6

Observations: clear milk top / clots bottom clear mixings

Reaction Type: Acid / Base neutralization / precipitation

Balanced Equation: prepared the clear and clots/says together.

Station #7

$$10.00 \text{ mL} \times 1.27 \text{ M} = 12.7 \text{ (mg OH)}$$

$$0.9573 \quad 12.1577$$

Station #8

It is used to measure the water and the funnel / breaker
other one is to control the gas

Station #9

$$75 \text{ mL} \times 3.43 \text{ M} = 257.89$$

Station #10

$$0.499 \text{ M} \quad 44.3437 \text{ g}$$

Extra Space

CHEM 3A - Lab Practical - VERSION C

Section #: _____

Name: Hailey AguirreStudent ID #: 0873290

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version C.

Station #1

$$50\text{mL} \times \frac{1\text{L}}{1000\text{mL}} = 0.05000\text{L}$$

$$[\text{M}]\text{ of NaCl} = \frac{5.0234\text{g}}{58.4427\text{g/mol}} \quad [\text{M}] = \frac{0.085956\text{mol}}{0.05000\text{L}} \quad \boxed{1.719\text{M}}$$

Station #2

$$\text{mass of aluminum foil} = 0.3302\text{g}$$

Station #3

$$\text{temp} = 21.99^\circ\text{C}$$

Station #4

$$q = m \times c \times \Delta T$$

$$q = 35.0392\text{g} \times 4.184\text{J/g}^\circ\text{C} \times 21.1^\circ\text{C}$$

$$q = 3093.34467$$

Station #5

$$20.0\text{mL}$$

Station #6

Observations: turned milky white color; started separating; looks almost

Reaction Type: redox

powdery on inside

Balanced Equation:

Station #7

Conc = 0.9573M

V = 10.00 mL

Initial = 1.27 mL

Final 11.27 mL

Station #8

10 mL pipet - used to measure liquids that are placed into beakers

Can't recall name - used to observe substances such as powders

Station #9

3.85 cm

Station #10

Mass of Mg used = 0.1495g

Mass of Mg_xO_y compound = 0.2479g

Mass of oxygen in Mg_xO_y compound = 0.0984g

What is the mass percent of magnesium in the Mg_xO_y sample?

0.1495g / 0.2479g

14.95%

Extra Space

CHEM 3A - Lab Practical - VERSION C

Section #: _____

Name: Pa FangStudent ID #: 0737921

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version C.

Station #1

50.00 mL of "solution A"
 weighed = 5.0234 g of NaCl
 molar mass of NaCl is 58.44277 g/mol
 $\text{molarity (M)} = (\text{mol solute}) / \text{L (solution)}$
 calculate the molarity of "solution A"

$$58.44277 - 5.0234 =$$

$$53.41937 \text{ g}$$

Station #2

Aluminum = 0.3301 g $2(0.3301 \text{ g}) + 2\text{KOH} + 4\text{H}_2\text{SO}_4 + 22\text{H}_2\text{O} \rightarrow 2(474.46) + 3\text{H}_2$
 $2\text{Al} + 2\text{KOH} + 4\text{H}_2\text{SO}_4 + 22\text{H}_2\text{O} \rightarrow 2\text{KAl(SO}_4)_2 + 12\text{H}_2\text{O} + 3\text{H}_2$
 Aluminum foil (Al) $\rightarrow$ 26.98 g/mol
 $\text{Alum (KAl(SO}_4)_2 \cdot 12\text{H}_2\text{O)} \rightarrow 474.46 \text{ g/mol}$
 from measure, what is the maximum amount of Alum (in grams) that could be made.

Station #3

temp. $\text{H}_2\text{O} = 21.1$ $P_{\text{total}} = P_{\text{H}_2} + P_{\text{H}_2\text{O}}$
 $P_{\text{total}} = 756.7 \text{ mmHg}$ $756.7 = 42 + 21.1$
 pressure of hydrogen gas = 42

Station #4

Calculate heat absorbed or released.
 $q = M \times C \times \Delta T$ $q = 35.0392 \text{ g} \times 4.184 \text{ J/g}^\circ\text{C} \times 21.1^\circ\text{C}$
 mass of NaOH added = 50.25 g
 mass total solution (m) = 35.0392 g
 specific heat of solution (c) = 4.184 J/g $^\circ\text{C}$
 change in temp. (ΔT) = 21.1 $^\circ\text{C}$
 Heat (q) ?

Station #5

Determine volume object "L" by displacement of water.

start = 20 mL $28 - 20 = 8 \text{ mL}$
 finish = 28 mL

Station #6

Observations: After mixing the two, it turned it a milky, cloudy mixture.

Reaction Type: Acid/Basic Neutralization.

Balanced Equation: $\text{BaCl}_2 + \text{Na}_3\text{PO}_4$

Station #7

Final buret reading = 19.0 mL

Station #8

Round glass → used to see experiments clearer.

6 mL → used for measuring liquids up to 6 mL

Station #9

Length of line = 3.90 cm

Station #10

$$\begin{array}{r} 0.2479\text{g} \\ - 0.1495\text{g} \\ \hline 0.0984\text{g} \end{array}$$

Extra Space

CHEM 3A - Lab Practical - VERSION B

Section #: _____

Name: Rita SarkisyanStudent ID #: 0891883

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version B.

Station #1

- 22.49
35.95
- 11.7479 flask + stopper (empty)
 - 70.1879

Station #2

- 7.2669
- mass of Alun: 27.1937g
 - It is theoretical yield

Station #3

- $$\begin{array}{r} 44.5 \\ - 21.5 \\ \hline 23 \end{array}$$

$$\frac{400}{23}$$
- volume of gas 1.7291g

Station #4

- Temperature went from 23.0°C to 21°C
- The reaction was endothermic

Station #5

- h 3.6 cm $(3.6)(3.6)(1.6) = 20.736 \text{ cm}$
 w 1.6 cm $(3.6)(3.6) = 12.96$
 L 3.6 cm Longest dimension is 3.6 cm

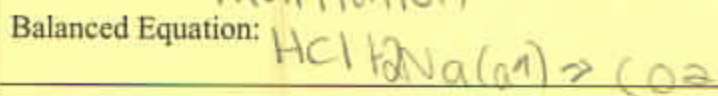
Station #6

Observations: once mixed together it changed to a yellow color for a second and fizzed up

Reaction Type:

Precipitation

Balanced Equation:



Station #7

$$\frac{10.00 \text{ mL}}{1.27 \text{ mL}} = 7.8740$$

$$\frac{10.00 \text{ mL}}{0.9573} = 0.9573$$

Station #8

- test tube - used for swirling any chemicals
- flask - used to pour chemicals

Station #9

- 28.5 - Graduated cylinder
- 25.00 - flask

Station #10

29.305

15.949

1.29.305

2.15.949

3.15.949

•

Extra Space

CHEM 3A – Lab Practical - VERSION CName: Acran Andrade

Section #: _____

Student ID #: 0604264

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version C.

Station #1

5.872 m

Station #2

1.88g

Station #3

22°C 756.6

$$14.827 + 756.6 = 776.427$$

Station #4

$$q = 35.0392 \times 4.184 \times 21.1$$

$$= 3093.34$$

Station #5

7.5 ml

Station #6

Observations: white cloudy substance formed

Reaction Type: precipitation

Balanced Equation: $2\text{HCl} + \text{BaPO}_4$

Station #7

20.10 ml

Station #8

water glass, Buret

Station #9

3.7 cm

Station #10

60.3 %

Extra Space

CHEM 3A – Lab Practical – **VERSION A**Section #: Tues 630-7:55pmName: Sabrina GarciaStudent ID #: 0490011

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version A**.

Station #1

$$18.2346g \quad \frac{10.00m}{18.2346g} = 1.82346g$$

Station #2

$$8.879g = \frac{7.4415g}{8.879g} \times 100 = 119.317g$$

Station #3

$$1.008 + 0.0475g = 1.0555g \quad \text{calculate the mass hydrogen gas in grams}$$

Station #4

$$20.3^{\circ}C$$

Station #5

$$V = l \times w \times h$$

$$\begin{aligned} &3.6cm \text{ L} \\ &1.2cm \text{ w} \\ &1.1cm \text{ h} \end{aligned}$$

$$V = 4.752cm^3$$

Station #6

Observations: milky/chalky

Reaction Type: Redox

Balanced Equation: $\text{CaCl}_2 \rightarrow \text{Na}_2\text{CO}_3$

Station #7

$$10.00 \text{ mL} - 1.27 = 8.73 \text{ mL}$$

$$0.9573 \text{ M}$$

Station #8

A: ^{Bunsen} Burner - to heat up liquids

A: Graduated cylinder - measure/pour liquids

Station #9

$$100.00 \text{ mL} - 2.5789 \text{ M} = 97.4211 - 25.00 \text{ mL} =$$

$$72.4211 \text{ mL}$$

Station #10

$$22.1715 - 21.8563 = 0.3155 \text{ g}$$

$$22.1722 - 21.8563 = 0.3159 \text{ g}$$

Extra Space

CHEM 3A - Lab Practical - VERSION A

Section #: _____

Name: Matthew PyreStudent ID #: 0523330

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version A.

Station #1

$$\begin{array}{rcl}
 10\text{mL} & 9.5130 & 6.2707 \\
 \rho = \frac{m}{V} & 15.7437 & 10\text{mL} \\
 & & \boxed{\rho = 0.62707\text{g/mL}}
 \end{array}$$

Station #2

$$\frac{7.4415\text{g}}{8.879\text{g}} \times 100\% = 83.8\%$$

$$\boxed{= 84\%}$$

Station #3

$$\begin{array}{rcl}
 0.0475\text{g Mg} & \text{H } 1.008 & 2 \\
 2.016 & 0.02375 & 2.016 \\
 & & \boxed{0.09576\text{g H}}
 \end{array}$$

Station #4

$$\boxed{22^\circ\text{C}}$$

Station #5

$$\begin{array}{l}
 V = l \times w \times h \\
 l = 3.8\text{cm} \\
 w = 1.3\text{cm} \\
 h = 1.3\text{cm} \\
 \boxed{V = 6.422}
 \end{array}$$

Station #6

Observations: Chemical turned cloudy white

Reaction Type: Precipitation

Balanced Equation: $\text{CaCl}_2 + \text{Na}_2\text{CO}_3 \rightarrow 2\text{NaCl} + \text{CaCO}_3$

Station #7

10.00mL

1.27mL

1.27mL - 25mL =

23.73mL

Station #8

Bunsen burner

25mL flask

Station #9

25.00mL of a 2.5789M

100.00mL

75.00mL

0.075

2.5789M

0.1

2.5789

0.0388M

Station #10

- 21.8563 — g

- 21.9751 — g

- 22.1715 — g (1st weighing)

- 22.1722 — g (2nd weighing)

0.0007 O reacted

Extra Space

CHEM 3A - Lab Practical - VERSION BSection #: 1Name: Rocella MAROUTZStudent ID #: 0345854

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose version B.

Station #1

$$100.1147g = \frac{46.9007g}{51.9244g} \times 100$$

$$= 98.7276g$$

$$\text{repp} =$$

$$\frac{98.8411g}{100.1147g} \times 100$$

Station #2

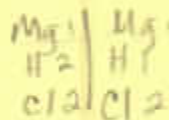
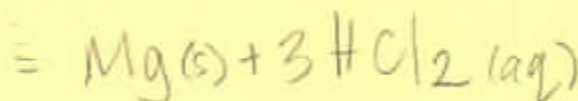
$$d = \frac{m}{v}$$

theoretical yield

$$d = \frac{7.4415g}{7.4415g} \quad d = 1$$

(el was confused as to where the numbers to the solution were)

Station #3



Station #4

The temperature went down. The reaction was exothermic.

Station #5

$$3.5 \text{ cm} = 3.5 \times 1000 = 3500 \text{ mm}$$

Station #6

Observations: clear, saw solutions mix briefly then clear

Reaction Type: gas

Balanced Equation: $1 \text{ mL Na}_2\text{CO}_3 + 1 \text{ mL HCl} = 2 \text{ Na}_2\text{CO}_3\text{HCl}$

Station #7

Down 10.00 mL

conc. 0.9573 M

initial - 1.27 mL

Final - 17.12 mL

$= 0.0007 \text{ mol}$ 873

Station #8

• test tube (pour initial solutions)

• flask (pour contents of test tube to see reactions/temp)

Station #9

28 mL - graduated cylinder

25 °C - flask

Station #10

mass 0.1524 g

Compound 0.2527 g

Oxygen 0.1003 g

% = 0.1663%

Extra Space